Strict negativity of a Chen–Larson hypergeometric coefficient

A computer-assisted and formally verified proof

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Abstract

Chen and Larson reduce a geometric vanishing problem for strata of holomorphic abelian differentials to the non-vanishing of a coefficient in a quotient of hypergeometric generating series. We prove the stronger statement that this coefficient is always strictly negative in the two residue classes covered by their Proposition 5.1. More precisely, let

$$\mathsf{C}(t) = \sum_{r \geq 0} \frac{(6r)!}{(3r)!(2r)!} \left(\frac{t}{72}\right)^r .$$

If $g \geq 2$, $g \equiv 0, 2 \pmod 3$, $a = \lfloor g/3 \rfloor + 1$, and $\mu = (m_1, \ldots, m_n)$ is a positive partition of $2g-2$, then

$$[t^a] \frac{\prod_{i=1}^n \mathsf{C}(t/(m_i+1))}{\mathsf{C}(t)^{2g-2+n}} < 0.$$

The proof first removes the partition by a coefficientwise majorization. A rational sign-lock estimate then shows that all convolution indices above $0.9a$ contribute nonpositively. The remaining positive part is controlled by a direct saddle method based on finite exponential prefixes, rather than on real-analytic exponential estimates. Small ranges are discharged by exact enumeration and verified rational or dyadic interval certificates; the large range is closed by symbolic ratio and reserve bounds. The complete argument has been formalized in Lean 4 and Mathlib. We also describe the role of long-horizon generative-AI assistance in the development and the precise trust boundary of the computational certificates.

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1 Introduction

1.1 Geometric context

Chen and Larson study tautological classes on strata of differentials. In Section 5 of their work [1], they pull back a relation among κ-classes on $\mathcal{M}_g$ to a stratum associated with a partition $\mu = (m_1, \ldots, m_n)$ of $2g-2$. For the residue classes $g \equiv 0, 2 \pmod 3$, their Proposition 5.1 reduces the desired vanishing to a single coefficient of the series

$$\frac{\prod_{i=1}^{n} \mathsf{C}(t/(m_i+1))}{\mathsf{C}(t)^{2g-2+n}}, \qquad \mathsf{C}(t) = \sum_{r \geq 0} \frac{(6r)!}{(3r)!(2r)!} \left(\frac{t}{72}\right)^r. \tag{1}$$

For a positive partition $\mu = (m_1, \ldots, m_n)$ of $2g-2$ — a tuple of integers $m_i \geq 1$ summing to $2g-2$ — let $\mathcal{P}(\mu)$ denote the projectivized stratum of holomorphic abelian differentials whose zero orders are $m_1, \ldots, m_n$, and let η denote its tautological line-bundle class, in the convention of Chen–Larson. Their Conjecture 1.4 predicts a nontrivial tautological relation in the first degree not excluded by their independence theorem, namely $a = \lfloor g/3 \rfloor + 1$; for $g \equiv 0, 2 \pmod 3$, Proposition 5.1 reduces the geometric vanishing $\eta^a = 0$ to the nonvanishing of the single coefficient in (1). The complementary class $g \equiv 1 \pmod 3$ is governed by their Proposition 5.2, which uses a different, derivative-corrected series, and is not addressed here. The geometric statement requires only that this coefficient be nonzero; our main result gives a uniform sign.

Theorem 1.1 (Main theorem)**.** *Let $g \geq 2$ satisfy $g \equiv 0$ or $2 \pmod 3$, and set*

$$a = \lfloor g/3 \rfloor + 1.$$

For every positive partition $\mu = (m_1, \dots, m_n)$ of $2g-2$,

$$[t^a] \frac{\prod_{i=1}^n \mathsf{C}(t/(m_i+1))}{\mathsf{C}(t)^{2g-2+n}} < 0. \tag{2}$$

In particular, the coefficient in Chen–Larson Proposition 5.1 is nonzero.

The strict negativity is stronger than the nonvanishing needed geometrically; we attach no geometric meaning to the sign itself.

Corollary 1.2. *In the setting of Chen–Larson Proposition 5.1, the class η^a vanishes for every positive holomorphic partition when $g \equiv 0, 2 \pmod 3$.*

Proof. Combine Theorem 1.1 with Chen–Larson Proposition 5.1. □

1.2 Outline

After the geometric reduction, the statement is no longer one of algebraic geometry: it is a uniform claim about a single coefficient of a quotient of power series. Two features make it hard. The quotient depends on an arbitrary partition $\mu = (m_1, \dots, m_n)$ of $2g-2$, and the same strict sign is required for every genus in the two residue classes under consideration and for every such partition. The proof meets these in turn: first it removes the partition, leaving an expression in two integer parameters; then it proves the sign for that expression uniformly — in the large-a argument, without expanding the coefficients themselves (in the finite range they are computed exactly). This subsection describes the argument in words; every object named here is defined precisely in the sections that follow.

Removing the partition. The one structural fact we need about C is that its logarithm has nonnegative coefficients: writing

$$\log \mathsf{C}(t) = \sum_{r \geq 1} c_r t^r,$$

one has $c_r > 0$ (proved in Section 3). Set $q_i = m_i + 1 \geq 2$ and $N = \sum_i q_i = 2g - 2 + n$; the coefficient of interest is $[t^a] \prod_i \mathsf{C}(t/q_i)/\mathsf{C}(t)^N$. Rescaling t inside the logarithm writes each factor of the numerator as an exponential, and the product as a single exponential,

$$\mathsf{C}(t/q_i) = \exp\Big(\sum_{r \geq 1} c_r\, q_i^{-r}\, t^r\Big), \qquad \prod_{i=1}^n \mathsf{C}(t/q_i) = \exp\Big(\sum_{r \geq 1} c_r\, \sigma_r\, t^r\Big), \quad \sigma_r := \sum_{i=1}^n q_i^{-r}.$$

The coefficients of an exponential $\exp(\sum_r a_r t^r)$ are polynomials in the a_r with nonnegative coefficients; since here $a_r = c_r \sigma_r \geq 0$, each coefficient of the product is a nondecreasing function of every σ_r. Now $q_i \geq 2$ gives $\sigma_r \leq n\, 2^{-r}$, and $n \leq N/2$, so $\sigma_r \leq \frac{N}{2}\, 2^{-r}$; replacing σ_r by this larger value turns the product into exactly $\mathsf{C}(t/2)^{N/2}$. Hence the numerator is dominated coefficientwise,

$$[t^r] \prod_i \mathsf{C}(t/q_i) \;\leq\; [t^r]\, \mathsf{C}(t/2)^{N/2} \qquad (r \geq 0),$$

by an upper bound that no longer remembers the partition. Substituting it for the numerator and dropping the convolution terms that are automatically nonpositive produces a partition-free upper bound $U_a(N)$ for the target coefficient — a *majorant* — whose explicit form is recorded in (8). It then remains to prove

$$U_a(N) < 0 \qquad \text{for the two integers } a, N \text{ subject to } 6a - 7 \leq N \leq 12a - 8.$$

A negative main term against small positive corrections. It is convenient to rescale the coefficients of the two relevant series. Let $X_r(N)$ be the normalized coefficient of t^r in the denominator $\mathsf{C}(t)^{-N}$, and $Y_r(N)$ the normalized coefficient of t^r in $\mathsf{C}(t/2)^{N/2}$ (the precise normalizations are fixed in Section 2). In these terms the majorant takes the shape

$$\frac{U_a(N)}{Nc_a} = \underbrace{X_a(N)}_{\text{one negative term}} + \text{(positive correction terms)}.$$

The proof is a contest between the two sides: we bound $X_a(N)$ above by a *negative* number with an explicit margin, and bound every positive correction below that margin. The corrections come from the convolution and are indexed by splittings $a = k + j$ with $1 \le k < a$.

Most splittings contribute nothing. We prove a *sign lock*: once $k > 0.9a$, the coefficient $X_k(N)$ is already negative throughout the whole range of N, so such terms only help an upper bound and may be dropped. This is the technical heart of the negative side, and it leaves only the splittings with $1 \le k \le \lfloor 0.9a \rfloor$ to be controlled.

Bounding coefficients without computing them. For the surviving corrections we never expand the coefficients as numbers — that would be hopeless for large a. Instead we use one elementary principle. If a series is the exponential of a series with nonnegative coefficients, its coefficient of t^m is bounded, for any positive rescaling of t, by a finite truncation of an ordinary exponential,

$$P_m(x) = \sum_{r=0}^{m} \frac{x^r}{r!}.$$

Choosing the rescaling well makes this bound sharp; this is the classical saddle-point idea, but carried out so that every quantity stays rational[1] and no limit, contour, or transcendental constant ever enters. Two finite exponentials multiply in the same spirit,

$$P_m(x)\, P_n(y) \le P_{m+n}(x+y) \qquad (x, y \ge 0),$$

which is the binomial theorem applied after enlarging a rectangle of indices to a triangle. This last inequality is more important than it looks: an early formalization attempt bounded the two factors $\mathsf{C}(t)^{-N}$ and $\mathsf{C}(t/2)^{N/2}$ separately and multiplied, which discarded a large power of 2 and produced a false target (Section 10). The proof that works keeps the product together and combines the finite exponentials only at the very end.

Finite checking and the infinite tail. The argument is split deliberately between explicit finite verification and uniform inequalities. For $a \le 8$ every partition is enumerated and the exact rational coefficient is checked. For $9 \le a \le 60$ the majorant $U_a(N)$ is checked exactly over the rectangle, and for $61 \le a \le 400$ the same check is run in verified interval arithmetic — a small calculator for intervals of rationals whose soundness is itself proved (so that rounding can never certify a false inequality). For all $a \ge 401$ the sign lock supplies the negative margin and the saddle bounds reduce the positive side to a tiny budget: the strip $401 \le a \le 2000$ is verified directly,

[1] This insistence on staying rational reflects the formal proof more than the mathematics. The Lean development works entirely over $\mathbb{Q}$: quantities that an informal asymptotic argument might estimate with Stirling's formula or the real exponential are instead bounded by finite rational inequalities, so the proof checker verifies everything by exact arithmetic with no real-analysis library and without formally introducing π or e. The slight loss in the resulting constants is absorbed by the large slack in the final budget.

and the strip $2001 \le a < 3000$ together with the genuinely infinite range $a \ge 3000$ are closed by showing that consecutive terms shrink by a fixed rational ratio, so each tail is at most its first term times a geometric series. The positive budget never exceeds 10^{-8}, while the negative margin is strictly larger; hence $U_a(N) < 0$, and the Chen–Larson coefficient is negative.

In one line, the proof is the chain

$$\begin{gathered}
\text{Chen–Larson coefficient } b_a(\mu) \\
\downarrow \text{ coefficientwise majorization} \\
\text{partition-free majorant } U_a(N) \\
\downarrow \text{ sign lock: } X_k(N) \le 0 \text{ for } k > 0.9a \\
\text{negative main term } + \text{ small positive corrections} \\
\downarrow \text{ joint finite-prefix saddle bound on } B_k Q_{a-k} \\
\text{explicit rational summand majorant} \\
\downarrow \text{ finite certificates } (a \le 2000) \text{ and ratio-controlled tails} \\
U_a(N) < 0.
\end{gathered}$$

1.3 Formal verification and scope

The theorem is formalized over $\mathbb{Q}$ in Lean 4 and Mathlib. The public interface names the hypergeometric series, proves its coefficient formula, characterizes the quotient generating series by the identity in (1), and exposes strict negativity and non-vanishing. The exact public theorem is reproduced in Appendix A.

The finitely many explicit computations — enumerations, exact rational checks, and interval certificates — are discharged by Lean's `native_decide`, which verifies a finite claim by compiling it to a program and running it. This makes the compiler part of what one trusts for those steps, in exchange for being able to check millions of cases. The trade is recorded openly: Lean's `#print axioms` command lists exactly which axioms a given theorem rests on, and for our final theorem that list is the standard logical axioms of Mathlib together with two named native-evaluation axioms and nothing else. Section 9 gives the precise trust boundary; the rest of the proof is checked by Lean's small logical kernel in the usual way.

2 The partition-free majorant

Let $\mu = (m_1, \dots, m_n)$ be a positive partition of $2g-2$ and put

$$q_i = m_i + 1 \ge 2, \qquad N = \sum_{i=1}^{n} q_i = 2g - 2 + n. \tag{3}$$

Set

$$b_a(\mu) = [t^a] \frac{\prod_i \mathsf{C}(t/q_i)}{\mathsf{C}(t)^N}. \tag{4}$$

For $g \equiv 0, 2 \pmod 3$ and $a = \lfloor g/3 \rfloor + 1$, one has

$$g = 3a - 3 \quad \text{or} \quad g = 3a - 1.$$

Since $1 \le n \le 2g - 2$, every relevant value of N lies in the enlarged rectangle

$$6a - 7 \le N \le 12a - 8. \tag{5}$$

The enlargement is convenient because it treats both residue classes at once.

Define

$$
\begin{aligned}
P_r(\mu) &= [t^r] \prod_i \mathsf{C}(t/q_i), \\
B_r(N) &= [t^r] \mathsf{C}(t)^{-N}, \\
Q_r(N) &= [t^r] \mathsf{C}(t/2)^{N/2}.
\end{aligned} \tag{6}
$$

Here rational powers are interpreted through the exponential of the formal logarithm. Since the logarithmic coefficients c_r are nonnegative, coefficients of

$$
\exp\left(\alpha \sum_{r \geq 1} c_r 2^{-r} t^r\right)
$$

are polynomials in α with nonnegative coefficients. Moreover, $q_i \geq 2$ and $n \leq N/2$. It follows that

$$
0 \leq P_r(\mu) \leq Q_r(N). \tag{7}
$$

Expanding (4), dropping nonpositive convolution terms, and using (7) gives the partition-free majorant

$$
b_a(\mu) \leq U_a(N) := B_a(N) + Q_a(N) + \sum_{\substack{1 \leq k < a \\ B_k(N) > 0}} B_k(N) Q_{a-k}(N). \tag{8}
$$

Thus Theorem 1.1 follows once we prove

$$
U_a(N) < 0 \tag{9}
$$

throughout (5), except for the finitely many values where we instead verify $b_a(\mu) < 0$ directly.

2.1 Normalized form

For $r \geq 1$ define

$$
X_r(N) = \frac{B_r(N)}{N c_r}, \qquad Y_r(N) = \frac{2^r Q_r(N)}{(N/2) c_r}, \tag{10}
$$

and set

$$
R_{k,a} = \frac{c_k c_{a-k}}{c_a}. \tag{11}
$$

The standard exponential-coefficient recurrence gives

$$
X_1 = -1, \qquad X_r = -1 - \frac{N}{r} \sum_{k=1}^{r-1} k R_{k,r} X_{r-k}, \tag{12}
$$

$$
Y_1 = 1, \qquad Y_r = 1 + \frac{N}{2r} \sum_{k=1}^{r-1} k R_{k,r} Y_{r-k}. \tag{13}
$$

A direct calculation transforms (8) into

$$
\frac{U_a(N)}{N c_a} = X_a(N) + 2^{-a-1} Y_a(N) + \frac{N}{2} \sum_{\substack{1 \leq k < a \\ X_k(N) > 0}} R_{k,a} 2^{-(a-k)} X_k(N) Y_{a-k}(N). \tag{14}
$$

The proof therefore consists of a uniform negative lower margin for $-X_a$ and an upper budget for the remaining two positive contributions.

3 Logarithmic coefficients

The hypergeometric series C satisfies a Riccati differential equation. Equating coefficients in the logarithmic derivative yields

$$c_1 = \frac{5}{6}, \qquad c_r = 6(r-1)c_{r-1} + \frac{6}{r}\sum_{i=1}^{r-2} i(r-1-i)c_i c_{r-1-i} \quad (r \geq 2). \tag{15}$$

In particular, $c_r > 0$. Write

$$c_r = 6^r (r-1)! d_r. \tag{16}$$

Then

$$d_r = d_{r-1} + \frac{1}{r}\sum_{i=1}^{r-2} \frac{d_i d_{r-1-i}}{\binom{r-1}{i}}. \tag{17}$$

The sequence d_r is nondecreasing and begins with $d_1 = d_2 = 5/36$.

The formal proof uses rational bounds throughout.

Lemma 3.1 (Uniform coefficient bounds)**.** *For every $r \geq 1$,*

$$\frac{5}{36} \leq d_r \leq \frac{4}{25}. \tag{18}$$

Consequently,

$$\frac{5}{36}6^r(r-1)! \leq c_r \leq \frac{4}{25}6^r(r-1)!. \tag{19}$$

Proof. The lower bound follows immediately from (17). For the upper bound, let $A_r = [t^r]\mathsf{C}(t)$ and $F_r = A_r/(6^r(r-1)!)$. One has $d_r \leq F_r$ and

$$\frac{F_{r+1}}{F_r} = 1 + \frac{5}{36r(r+1)}.$$

A rational telescoping argument bounds F_r by

$$\frac{F_3}{1-(5/36)(1/3-1/r)} \leq F_3\frac{108}{103} < \frac{4}{25}.$$

No estimate involving π is required. □

The reciprocal-binomial estimate

$$\sum_{i=1}^{p-1} \binom{p}{i}^{-1} \leq \frac{4}{p} \tag{20}$$

combines with (17) and (18) to give an increment bound.

Lemma 3.2 (Increment and ratio control)**.** *For $r \geq 2$,*

$$d_r - d_{r-1} \leq \frac{64/625}{r(r-1)}. \tag{21}$$

If $1 \leq s < m$, then

$$1 - \frac{2304}{3125}\frac{s}{m(m-s)} \leq \frac{d_{m-s}}{d_m} \leq 1. \tag{22}$$

Proof. The nonlinear summand in (17) is at most $(16/625)\binom{r-1}{i}^{-1}$. Summing with (20) gives (21). Telescoping from $m-s$ to m yields

$$d_m - d_{m-s} \le \frac{64}{625}\left(\frac{1}{m-s} - \frac{1}{m}\right).$$

Divide by $d_m \ge 5/36$. The upper ratio bound follows from monotonicity. □

A further consequence, used throughout the positive-part estimate, is

$$R_{k,a} \le \frac{576}{3125}\frac{1}{(a-1)\binom{a-2}{k-1}} \qquad (1 \le k < a). \tag{23}$$

Indeed, the d-factor contributes at most $(4/25)^2/(5/36) = 576/3125$, and the factorial quotient is exactly the reciprocal denominator in (23).

4 Finite verification through $a = 400$

The majorant is not strong enough for the smallest values of a, so the formal proof begins with direct finite verification.

4.1 Positive partitions for $a \le 8$

For $g \le 23$, equivalently $a \le 8$ in the relevant residue classes, every positive partition of $2g-2$ is generated and the exact rational coefficient $b_a(\mu)$ is evaluated. The largest enumeration is $p(44) = 75175$ partitions at $g = 23$. The executable theorem checks strict negativity for all of them. Independent cardinality statements verify selected partition counts, reducing the risk of an incomplete generator.

4.2 Exact majorant certificate for $9 \le a \le 60$

For $9 \le a \le 60$, the recurrence tables for c_r, $B_r(N)$, and $Q_r(N)$ are evaluated exactly over $\mathbb{Q}$. This verifies (9) at all 10764 pairs in the rectangle. As a regression anchor, the least-negative corner is recorded exactly as

$$\frac{U_9(100)}{100c_9} = -\frac{13391635371339739}{213818836571652096}. \tag{24}$$

4.3 Verified dyadic intervals for $61 \le a \le 400$

The remaining 470220 pairs are certified by a small executable interval kernel. A dyadic interval stores outward-rounded endpoints with a 192-bit mantissa. The exact rational recurrences are mirrored term by term, and separate soundness lemmas prove that the tables enclose the exact values. For the conditionally included term in (8), the checker uses the convex hull of the product interval and $\{0\}$; consequently it never needs to decide the sign of $B_k(N)$ from an interval approximation.

The interval kernel contains definitions only and imports no Mathlib code; its soundness theory is proved separately over $\mathbb{Q}$. Eight native-evaluation chunks cover the N-range. Their conjunction proves (9) for all $61 \le a \le 400$. The 192-bit precision matches the reference Arb calculation used during discovery, but Arb is not part of the trusted proof: Lean checks its own dyadic kernel and its rational enclosure semantics.

5 The effective sign lock

For $a \geq 401$, the leading term $X_a(N)$ is controlled analytically. More generally, the same estimate forces all sufficiently high convolution indices to be nonpositive.

Define

$$\zeta = \frac{5N}{36m}. \tag{25}$$

If $N \leq (40/3)m$, then $0 \leq \zeta \leq 50/27$. The recurrence for X_m is reorganized into a truncated alternating Poisson-type sum, a controlled near-range error, and a far signed tail. The pure alternating base is controlled by a twelve-term prefix together with a paired tail; on $0 \leq \zeta \leq 50/27$ its final even–odd pair ($r = 10, 11$) is nonnegative, so the first ten terms already supply the convenient uniform lower bound

$$\lambda = \sum_{r=0}^{9} \frac{(-50/27)^r}{r!} = \frac{678107852315029}{4323713773987629}. \tag{26}$$

The error analysis is organized into seven independent rational budgets:

source	allowance in the numerator of m^{-2}
factorial-ratio residual	426
d-ratio drift	13
two-block recentering	184
higher two-block terms	234
three-or-more blocks	573
cross terms	784
far tails	1
total	2215

Every entry is a theorem over rational numbers. The twelve-term prefix inequalities on $0 \leq \zeta \leq 50/27$ are certified after rescaling to $[0, 1]$ by nonnegative Bernstein coefficients. The remaining alternating tail is paired into nonnegative even–odd blocks.

Theorem 5.1 (Sign lock)**.** *If $m \geq 361$, $N \geq 1$, and $N \leq (40/3)m$, then*

$$X_m(N) \leq -\left(\lambda\left(1 - \frac{2}{m}\right) - \frac{2215}{m^2}\right) < 0. \tag{27}$$

Proof. Set $\zeta = 5N/(36m)$ and $p = m - s$. Expanding $B_m(N) = [t^m]\mathsf{C}(t)^{-N}$ by the s-fold contribution of the linear part of $-N \log \mathsf{C}$ and normalizing by Nc_m gives the finite sum

$$-X_m(N) = \sum_{s=0}^{m} \frac{(-\zeta)^s}{s!}\,\Pi_s\,D_s\,(1 + \varepsilon_{m-s}), \qquad \Pi_s = \prod_{i=1}^{s} \frac{1}{1 - i/m}, \quad D_s = \frac{d_{m-s}}{d_m}, \tag{28}$$

where $\varepsilon_p = E_p^-(N)/(-Nc_p) - 1$ is the normalized nonlinear residual, which the envelope of Section 6 bounds by $|\varepsilon_{m-s}| \leq 66/(5m)$ for $s \leq m/3$. Write $e_1(s) = s(s+1)/2$ and split each factor as

$$\Pi_s D_s (1 + \varepsilon_{m-s}) = 1 + \frac{e_1(s)}{m} - \frac{\zeta}{m} + w_s.$$

The linear part sums in closed form to the base

$$\mathcal{B}_m(N) := \sum_{s\geq 0} \frac{(-\zeta)^s}{s!}\Big(1 + \frac{e_1(s)}{m} - \frac{\zeta}{m}\Big) = \mathrm{e}^{-\zeta}\Big(1 - \frac{2\zeta - \zeta^2/2}{m}\Big) \geq \mathrm{e}^{-\zeta}\Big(1 - \frac{2}{m}\Big) \geq \lambda\Big(1 - \frac{2}{m}\Big),$$

valid for $0 \leq \zeta \leq 50/27$, the last step using the ten-term alternating lower bound $\mathrm{e}^{-\zeta} \geq \lambda$ of (26). The remaining error

$$\mathcal{E}_m(N) = \sum_{0\leq s\leq m/3} \frac{(-\zeta)^s}{s!} w_s + (\text{far tail}),$$

collects exactly the seven sources tabulated above — the Π_s gamma-product residual, the D_s drift, the two- and higher-block nonlinear terms of ε, their cross terms, and the $s > m/3$ tail. Each row of that table is the corresponding rational bound, and they sum to $|\mathcal{E}_m(N)| \leq 2215/m^2$. Finally $m^2(\lambda(1-2/m)) > 2215$ at $m = 361$ is an exact rational check, and $m^2\lambda(1-2/m)$ is increasing in m, so (27) holds for every $m \geq 361$. □

Let

$$k_{\max}(a) = \lfloor 9a/10 \rfloor .$$

If $a \geq 401$ and $k > k_{\max}(a)$, then $k \geq 361$ and

$$N \leq 12a - 8 \leq \frac{40}{3}k.$$

Theorem 5.1 therefore implies $X_k(N) \leq 0$. Hence the positive sum in (14) can be restricted to

$$1 \leq k \leq \lfloor 9a/10 \rfloor . \tag{29}$$

For these retained indices, $j = a - k$ satisfies $j \geq a/10$.

6 Finite exponential prefixes and the direct saddle method

The formal proof does not estimate the relevant coefficients by invoking the real exponential directly. Instead, it works with finite prefixes

$$P_m(x) = \sum_{r=0}^{m} \frac{x^r}{r!}. \tag{30}$$

This keeps all intermediate statements rational.

6.1 Generic finite saddle lemmas

Let $L_r \geq 0$, $L_0 = 0$, and write

$$E_m(L) = [t^m] \exp\left(\sum_{r\geq 1} L_r t^r\right).$$

For $\rho \geq 0$, scaling gives

$$E_m(r \mapsto \rho^r L_r) = \rho^m E_m(L).$$

Expanding the exponential and bounding the coefficient of a power by the corresponding total gas yields the finite saddle inequality

$$\rho^m E_m(L) \le P_m\left(\sum_{r=1}^{m} L_r \rho^r\right). \tag{31}$$

The other essential identity is the finite convolution bound.

Lemma 6.1 (Prefix convolution)**.** *For $x, y \ge 0$,*

$$P_m(x)P_n(y) \le P_{m+n}(x+y). \tag{32}$$

Proof. Expand the left side over the rectangle $0 \le p \le m$, $0 \le q \le n$, enlarge to the triangle $p+q \le m+n$, and group by $r = p+q$. The binomial identity gives

$$\sum_{p+q=r} \frac{x^p y^q}{p!q!} = \frac{(x+y)^r}{r!}.$$

All omitted summands are nonnegative. □

The same binomial argument proves

$$(1+d)^n \le P_n(nd) \qquad (d \ge 0) \tag{33}$$

and the elementary absorption

$$\frac{x^r}{r!} \le P_m(x) \qquad (0 \le r \le m,\ x \ge 0). \tag{34}$$

6.2 Factorial-gas estimates

The rational saddle radius is

$$\beta = \frac{34}{15}. \tag{35}$$

For $n \ge 40$ define

$$u_{n,r} = \beta^r \frac{(r-1)!}{n^r}.$$

Then

$$\frac{u_{n,r+1}}{u_{n,r}} = \frac{34r}{15n}. \tag{36}$$

Thus the sequence decreases and then increases (its consecutive ratio crosses 1 exactly once). Every term with $4 \le r \le n$ is bounded by $u_{n,4} + u_{n,n}$. The low and endpoint estimates formalized in the proof are

$$n(u_{n,2} + u_{n,3}) \le \frac{1}{6}, \qquad n^2 u_{n,4} \le \frac{1}{8}, \qquad n^2 u_{n,n} \le \frac{1}{2}. \tag{37}$$

For the last inequality, set

$$W_n = n^2 u_{n,n} = \beta^n \frac{n!}{n^{n-1}}.$$

An exact check gives $W_{40} \le 1/2$, and

$$\frac{W_{n+1}}{W_n} = \frac{\beta}{(1+1/n)^{n-1}} \le 1 \qquad (n \ge 40).$$

Combining (36)–(37) gives

$$\sum_{r=2}^{n} \beta^r \frac{(r-1)!}{n^r} \le \frac{1}{n}. \tag{38}$$

The proof has slack: after multiplication by n, its three budgets total $1/6 + 1/8 + 1/2 = 19/24$.

For the small branch, if $s \ge 32$ and $2 \le k \le s$, monotonicity of $(r-1)!/s^r$ after the first few terms gives

$$\sum_{r=2}^{k} \frac{(r-1)!}{s^r} \le \frac{1}{s^2} + \frac{8}{s^3} \le \frac{5}{4s^2}. \tag{39}$$

6.3 Coefficient saddle bounds

Let $B_k^+(N)$ denote the coefficient obtained from $\exp(N \sum_{r\ge1} c_r t^r)$; in particular $B_k(N) \le B_k^+(N)$. Take $s = \lceil \sqrt{N} \rceil$. Applying (31), (19), and (39) with $\rho = (6s)^{-1}$ yields

$$B_k^+(N) \le (6s)^k P_k\left(\frac{5s}{36} + \frac{1}{5}\right) \qquad (2 \le k \le s). \tag{40}$$

For the tempered branch, use

$$\rho_B = \frac{17}{45k} = \frac{\beta}{6k}, \qquad \rho_Q = \frac{34}{45j} = \frac{\beta}{3j}.$$

The gas estimate (38) gives

$$B_k^+(N) \le \left(\frac{45k}{17}\right)^k P_k\left(\frac{17N}{54k} + \frac{4N}{25k}\right), \tag{41}$$

$$Q_j(N) \le \left(\frac{45j}{34}\right)^j P_j\left(\frac{17N}{108j} + \frac{2N}{25j}\right). \tag{42}$$

The normalization uses the lower bound in (19) and the factorial inequality

$$\left(\frac{25r}{68}\right)^r \le r!, \tag{43}$$

proved by induction on r from the rational bound $(1+1/r)^r \le 68/25$, a rational replacement for $(1+1/r)^r < \mathrm{e}$. The choice (35) is adapted to this estimate because

$$\frac{15}{34}\frac{68}{25} = \frac{6}{5}.$$

Consequently the residual powers are absorbed by (33).

Theorem 6.2 (Direct product bounds)**.** *Let $j = a - k$ (so $k + j = a$) and $N \le 12a$, and recall $Q_j(N) \ge 0$.*

If $s \ge 32$, $2 \le k \le s$, $N \le s^2$, and $j \ge 40$, then

$$\begin{aligned} 2\,2^j B_k(N) Q_j(N) \le{}& \frac{2592}{25} kjc_kc_j \\ &\times P_{2a}\left(\frac{41}{36}s + \frac{j}{5} + \frac{29}{10}\frac{a}{j} + \frac{1}{5}\right). \end{aligned} \tag{44}$$

If $k, j \geq 40$, then

$$2\,2^j B_k(N)Q_j(N) \leq \frac{2592}{25} kjc_kc_j \times P_{2a}\left(\frac{a}{5} + \frac{57}{10}\frac{a}{k} + \frac{29}{10}\frac{a}{j}\right). \tag{45}$$

Proof. For (44), combine (40) and (42), normalize by c_kc_j, absorb $s^k/k!$ and $(6/5)^j$ using (34) and (33), and apply Lemma 6.1 repeatedly. The Q-gas satisfies

$$\frac{17N}{108j} + \frac{2N}{25j} \leq \frac{29}{10}\frac{a}{j}$$

because $N \leq 12a$. The coefficient of s is $1 + 5/36 = 41/36$.

For (45), combine (41) and (42). Since $k + j = a$, both residual powers combine to $(6/5)^a$, which is absorbed into $P_a(a/5)$. The gas estimates

$$\frac{17N}{54k} + \frac{4N}{25k} \leq \frac{57}{10}\frac{a}{k}, \qquad \frac{17N}{108j} + \frac{2N}{25j} \leq \frac{29}{10}\frac{a}{j}$$

complete the proof. The product of the two normalization constants is $(36/5)(72/5) = 2592/25$. $\square$

7 The positive budget for $a \geq 401$

We now insert Theorem 6.2 into the normalized majorant. Let

$$N_- = 6a - 7, \qquad N_+ = 12a - 8, \qquad s_- = \left\lceil \sqrt{N_-} \right\rceil, \qquad s_+ = \left\lceil \sqrt{N_+} \right\rceil, \qquad j = a - k.$$

For a rational $x < T$, define the certified exponential upper bound

$$\mathsf{E}_T(x) = \sum_{r=0}^{T-1} \frac{x^r}{r!} + \frac{x^T}{T!}\frac{1}{1 - x/T}. \tag{46}$$

The last term is a geometric majorant for the remaining exponential tail.

For $401 \leq a \leq 2000$, define

$$E_s(a,k) = \frac{1139}{1000}s_+ + \frac{j}{5} + \frac{29}{10}\frac{a}{j} + 1, \tag{47}$$

$$E_t(a,k) = \frac{a}{5} + \frac{57}{10}\frac{a}{k} + \frac{29}{10}\frac{a}{j} + 2. \tag{48}$$

For $401 \leq a \leq 2000$ and every retained k one has $0 \leq E_s(a,k), E_t(a,k) < 800$, so the truncated bound E_{800} of (46) is applicable (its geometric tail term needs the argument below the cutoff). The direct product theorem and elementary rational absorptions give

$$X_k(N)Y_j(N) \leq \frac{2581}{20}\frac{kj}{NN_+}\mathsf{E}_{800}(E_s(a,k)), \qquad k \leq \left\lceil \sqrt{N} \right\rceil, \tag{49}$$

$$X_k(N)Y_j(N) \leq \frac{2117}{20}\frac{kj}{N^2}\mathsf{E}_{800}(E_t(a,k)), \qquad k > \left\lceil \sqrt{N} \right\rceil. \tag{50}$$

For the small branch, the comparison

$$\frac{2592}{25}N_+ \leq \frac{5}{3}\frac{2581}{20}N$$

uses the lower edge $N \geq N_-$; the factor $5/3$ is absorbed as $P_1(2/3)$. The numerical coefficient $1139/1000$ exceeds $41/36$ by $1/9000$. For the tempered branch, $2592/25 < 2117/20$, and the extra $+2$ in (48) provides ample argument slack.

Combining (23), (49), and (50), the retained positive convolution is bounded by a sum of explicit terms

$$M_s(a,k) = \frac{65}{N_+}\frac{kj}{(a-1)\binom{a-2}{k-1}}2^{-j}\mathsf{E}_{800}(E_s(a,k)), \tag{51}$$

$$M_t(a,k) = \frac{96}{N_-}\frac{kj}{(a-1)\binom{a-2}{k-1}}2^{-j}\mathsf{E}_{800}(E_t(a,k)). \tag{52}$$

The branch union must respect the regime conditions: the small bound (49) applies only for $k \leq \lceil\sqrt{N}\rceil$ and the tempered bound (50) only for $k > \lceil\sqrt{N}\rceil$, and $\lceil\sqrt{N}\rceil$ moves with N across $s_- \leq \lceil\sqrt{N}\rceil \leq s_+$. Outside its regime M_t has no content and can be large, so one may not simply take the unconditional maximum. The theorem-facing edge term is therefore the regime-aware

$$M_{\mathrm{edge}}(a,k) = \begin{cases} M_s(a,k), & k \leq s_-, \\ \max\{M_s(a,k),\, M_t(a,k)\}, & s_- < k \leq s_+, \\ M_t(a,k), & s_+ < k, \end{cases} \tag{53}$$

which dominates the retained summand for every admissible N: if $k \leq \lceil\sqrt{N}\rceil$ then $k \leq s_+$ and the small bound, maximized at the upper edge $N = N_+$, applies; if $k > \lceil\sqrt{N}\rceil$ then $s_- < k$ and the tempered bound, whose prefactor is maximized at the lower edge $N = N_-$, applies; only in the overlap $s_- < k \leq s_+$ can the applicable branch depend on N, and there the maximum of the two is taken.

7.1 The finite strip $401 \leq a \leq 2000$

The formal certificate proves

$$\sum_{k \text{ retained}} M_{\mathrm{edge}}(a,k) \leq \frac{1}{200000000} \tag{54}$$

for every $401 \leq a \leq 2000$. The calculation is split into sixteen consecutive blocks of one hundred rows. Each block evaluates a verified upward-rounded fixed-point upper bound for (46) at scale 10^{20}. The executable Boolean is connected to the rational inequality (54) by a proved soundness theorem. Thus the displayed scan is not an external numerical assumption; it is the paper-level description of the Lean certificate interface. The prose reductions in Sections 2–7 are meant to explain the mathematics, while the Lean development is authoritative for the final constants and case splits.

The solo term in (14) is handled separately. The formal proof uses a sharpened factorial-composition estimate and proves

$$2^{-a-1}Y_a(N) \leq \frac{1}{200000000} \tag{55}$$

throughout the same rectangle. This is the corrected theorem-facing bound; it does not use the obsolete shortcut $\exp(-0.49a)$ from an earlier draft. Thus the total positive envelope is at most

$$\frac{1}{200000000} + \frac{1}{200000000} = 10^{-8}. \tag{56}$$

7.2 The prefix strip $2001 \le a < 3000$

The same product inequalities are proved pointwise in this strip, with cutoffs a and $8a$ in the rational exponential-tail bounds. Generated prefix ratio and raw-budget certificates cover the finite lower-tempered transition that is inconvenient for a single asymptotic monotonicity argument. The solo estimate is supplied by the same sharp rational factorial-composition route used in the tail. No grid of exact $B_k(N)Q_j(N)$ products is expanded.

7.3 The analytic tail $a \ge 3000$

For the infinite range, the direct saddle bounds imply raw denominator-cleared versions of (49) and (50), with constants 130 and 192. The reciprocal binomial factor is controlled by

$$\binom{n}{k} \ge \left(\frac{n}{k}\right)^k, \tag{57}$$

and by a rational entropy-shadow inequality derived from the maximal weighted binomial term. The retained range is divided into a small branch and lower, middle, and upper tempered bands. Within each band, adjacent-term quotients are bounded by explicit rational numbers below one, and the first or last term of the band is multiplied by the corresponding geometric reserve $1/(1-\varrho)$. The concrete constants built into the formal tail are the following.

band	direction	adjacent-term ratio ϱ	boundary term initialized
small ($k \le \lceil\sqrt{N}\rceil$)	forward	$\le \frac{1}{2}$	first term ($k = 1$)
tempered, lower/middle	forward	$\le \dfrac{4a-1}{4a}$	first term of the band
tempered, upper	reverse	$\le \dfrac{4a-1}{4a}$	last term ($k = \lfloor 0.9a \rfloor$)

The tempered range is split at the natural-number threshold $k = \lfloor a/3 \rfloor + 10$, and each band's geometric reserve is allocated inside the half-budget $\frac{1}{2} \cdot 10^{-8}$. The solo term is separately bounded through a $(10/7)^a$ envelope, as described below.

The lower-tempered prefix $2001 \le a < 3000$ is represented by generated quotient and raw-budget chunks; from $a = 3000$ onward, a symbolic sharp top-offset quotient theorem applies. The upper band is treated in the reverse direction. Separate unit-reserve theorems prove that the geometric multipliers fit inside the allocated half-budget. This hybrid construction proves the same uniform estimate as (54) for every $a > 2000$.

The solo term is bounded by splitting a closed factorial-composition sum into low-degree and remainder blocks, proving upper-edge monotonicity in N, and comparing the result to a $(10/7)^a$ envelope. The remaining dyadic factor then yields (55). All steps are rational inequalities; no floating-point assertion is imported into the theorem.

We summarize the outcome.

Theorem 7.1 (Positive envelope)**.** *For every $a \ge 401$ and every N satisfying* (5)*,*

$$2^{-a-1}Y_a(N) + \frac{N}{2} \sum_{\substack{1 \le k < a \\ X_k(N) > 0}} R_{k,a} 2^{-(a-k)} X_k(N) Y_{a-k}(N) \le 10^{-8}. \tag{58}$$

8 Completion of the proof

For $a \geq 401$, the rectangle gives $N \leq 12a - 8 < (40/3)a$, so Theorem 5.1, applied with $m = a$, yields

$$X_a(N) \leq -\left(\lambda\left(1 - \frac{2}{a}\right) - \frac{2215}{a^2}\right). \tag{59}$$

An exact endpoint comparison at $a = 401$, followed by monotonicity, proves

$$10^{-8} < \lambda\left(1 - \frac{2}{a}\right) - \frac{2215}{a^2}. \tag{60}$$

Combining (14), Theorem 7.1, and (60) gives $U_a(N) < 0$ for every $a \geq 401$.

Together with the direct partition enumeration for $a \leq 8$, the exact majorant certificate for $9 \leq a \leq 60$, and the dyadic interval certificate for $61 \leq a \leq 400$, this proves Theorem 1.1.

9 Formalization architecture and trust

9.1 Public statement layer

The repository now has a concise public facade. Its mathematical content is:

1. the definition of the hypergeometric power series C;
2. the coefficient identity for C;
3. the quotient-series identity

$$\mathsf{C}(t)^N B_\mu(t) = \prod_i \mathsf{C}(t/(m_i + 1));$$

4. strict negativity of the coefficient in Theorem 1.1;
5. the non-vanishing corollary.

The bridge from the recurrence-defined c_r to the formal power series is proved by showing that both C and $\exp(\sum_{r\geq 1} c_r t^r)$ solve the same formal differential equation with constant term one.

9.2 Proof layers

The production proof has the following conceptual dependency chain:

layer	role
series bridge	identifies the recurrence-defined coefficients with the Chen–Larson generating series
majorant	replaces the partition-dependent numerator by $Q_r(N)$ and proves $b_a(\mu) \leq U_a(N)$
finite certificates	handles $a \leq 400$ by enumeration, exact rationals, and verified dyadic intervals
sign lock	eliminates $k > \lfloor 0.9a \rfloor$ and supplies the negative margin
direct saddle	proves the finite-prefix product inequalities (44) and (45)
positive assembly	checks the bounded edge budget and closes the prefix and infinite tail by ratio/reserve estimates
completion	combines all ranges into the unconditional public theorem

9.3 Trust boundary

Almost all of the argument is checked by Lean's small logical kernel, the same trusted core that underlies every Mathlib proof. The exception is the family of large finite computations — the enumerations and the exact and interval certificates — which are run by `native_decide`: Lean compiles the finite Boolean check to machine code, executes it, and accepts the result. This is faster by orders of magnitude than kernel reduction and is what makes the millions of certificate cases feasible, but it widens the trusted base to include the compiler for those steps. Following the Lean reference manual [3], each such use is recorded by a named axiom, and for the present development the complete list of extra-logical assumptions is exactly two: `Lean.ofReduceBool` and `Lean.trustCompiler`. Nothing else — in particular no `sorry`, no project-specific axiom, and no floating-point assumption — enters the final theorem.

This is verifiable in one command. The repository ships a short audit script that prints, for the series bridge, the majorant inequality, the final negativity theorem, and the non-vanishing corollary, the full list of axioms each depends on; a reader can reproduce it from a clean checkout with

```
lake exe cache get   # download the prebuilt Mathlib
lake build           # check every proof, including the certificates
lake env lean scripts/PublicAxiomsReport.lean   # print the axiom lists
```

The completed proof and the compact public facade are preserved as tagged releases in the repository. As an additional audit, the external checker `lean4checker` can replay the ordinary (non-native) proof objects and can detect certain bugs involving import state or declaration processing. It does not, however, remove the compiler trust attached to the native-evaluation certificates: proofs that depend on `Lean.trustCompiler` cannot at present be validated by an external proof checker without first replacing those computations by kernel-checkable certificates [3].

10 Background and use of generative artificial intelligence

This project was carried out almost entirely through generative AI systems, and the way they were used is part of what the paper reports. It began after Hannah Larson's talk *Tautological classes on the strata of differentials* in the Algebraic Geometry and Moduli seminar at ETH Zurich on 10 June 2026, which described the conjecture of Chen and Larson. The author was in the audience and decided to attack the resulting power-series problem with general-purpose AI tools. The human role throughout was orchestration rather than mathematics: choosing the problem, designing and relaying prompts, running and maintaining the computational and Lean environment, deciding from high-level signals when an autonomous run had stalled and an outside review should be solicited, and taking final responsibility for this text. The author did not at any point analyze or contribute to the mathematical substance of the proof. The proof search, much of the formal development, and parts of this write-up were produced with substantial AI assistance; no theorem-level step in the final argument was accepted until it had been formalized in Lean or independently checked as mathematics.

We document this as a chronological sequence of *human–AI interaction cards*, following the format proposed by Feng et al. [5]. Each card states the task, the model, the essential prompt, a one-line summary of the reply, and the human action it led to; the card title links to the public conversation where one exists. The prompts are reproduced verbatim or in close paraphrase, because the prompting strategy is the part of the method most likely to transfer to other problems; the AI replies are only summarized, with the most relevant exchanges also archived alongside the source.

In the taxonomy of [5] we classify the project as **Level A**, essentially autonomous. The core mathematical content — the proof strategy, the analytic estimates, the entire Lean formalization, and the two diagnoses that redirected the search — was generated by AI systems, with no essential human mathematical contribution. The two interventions recorded in Cards 6 and 7 are not exceptions to this: each was a diagnosis produced by a *separate* AI model, which the author solicited on the basis of a high-level signal (an autonomous run that had continued for two days without converging) and relayed to the worker, without examining the mathematics himself. This is precisely what separates Level A from the collaborative Level C: the orchestration was human, but the mathematics was not. As to significance, the theorem settles the coefficient nonvanishing that Chen and Larson isolate as their Proposition 5.1 — a step toward their Conjecture 1.4 that they, and colleagues with whom they discussed it, had left without a known route — which the author calibrates around Levels 1–2 of [5], offered for and not in place of expert review. In every case it is the Lean proof, not its provenance, that certifies the theorem.

The development had a simple shape, worth stating before the details. A single ChatGPT 5.5 Pro conversation (Card 1) was the central strand in which the mathematics was worked out and successively written up as LaTeX and Python. Into that strand the author repeatedly fed two distinct kinds of input from separate conversations — adversarial *audits* to surface errors (Card 2) and strategic *guidance* from a second model to suggest pivots (Card 3) — interleaved with simple requests to continue, producing ten drafts before formalization began. The formalization then moved into Lean, first broadly (Card 4) and then under a long-horizon autonomous goal (Card 5), with two moments (Cards 6 and 7) at which the author, prompted by a high-level signal rather than by the mathematics, solicited an outside AI review and relayed its diagnosis to redirect the search.

Card 1: Central investigation strand **ChatGPT 5.5 Pro**

Task. Attack the Chen–Larson coefficient problem from scratch and develop a candidate proof.

Prompt.

> *Can you have a look at Conjecture 1.4 of* arXiv:2603.23850 *(and its explicit reformulation Proposition 5.1) and make a serious, sustained, high-effort research attack at trying to crack it? This is an open problem, but in similar conversations you already managed to crack several such problems, and you might have a perspective on this power series that is not available easily to the authors. Would be really curious if you can make progress here!*

Reply (summary). Proposed the coefficientwise majorization that removes the partition, the normalized coefficients X_r, Y_r, and the strategy of a negative leading term against small positive corrections.

Continuation. This single conversation became the central strand: in later turns it produced the LaTeX write-up and Python verification scripts, and carried the mathematics through ten successive drafts, fed by the audits and guidance of Cards 2 and 3.

Card 2: Adversarial mathematical audit **ChatGPT 5.5 Pro**

Task. Surface every defect in a draft package (notes and code) before trusting it.
Prompt.

> *Consider the appended zip file. Please read the contained mathematical notes in detail and inspect all relevant code files, then proceed with a full mathematical audit, going paragraph by paragraph to identify any mathematical errors, gaps in arguments, hand-waving, misapplications of known results, or mis-cited theorems — anything that could affect the validity of the presented arguments. Then give me a full report.*

Reply (summary). Returned concrete defects — a far-tail constant valid only in a smaller range, a positive-part bound vacuous at $a = 401$, a single-edge scan that missed the worst value of N — each then repaired.
Human action. Fed each reported defect back into the central strand for repair, and re-ran the audit on the next draft.
Technique. A fresh conversation used purely adversarially: an exhaustive, paragraph-level audit of notes and *code, repeated each revision.*

Card 3: Strategic guidance **Claude Fable 5**

Task. Get higher-level direction, not line edits, when a draft stalled.
Prompt.

> *I wanted to ask for advice about the appended pure mathematics research note. For the theorem it tries to prove it shows some partial progress, but gaps remain. Please look through it carefully, think hard, do your own explorations or search for sources, and then give me a report focused on the most helpful advice possible for a second AI which will be tasked to continue pushing for a solution — concrete pointers to results, obstructions, a higher-level pivot, anything that could help crack the problem.*

Reply (summary). Returned strategic guidance — which effective-constant program to pursue, the worst-case regimes to target, and higher-level pivots — phrased as instructions the authoring model could act on.
Human action. Forwarded the guidance into the central strand to shape the next revision.
Technique. A second model used not to find errors but to suggest direction, with the prompt explicitly asking for advice aimed at the next AI in the loop.

Card 4: First formalization push **Claude Fable 5 (Claude Code)**

Task. Begin the Lean development: the statement layer, the series bridge, the partition-free majorant, the finite certificates, and the sign lock.
Prompt (excerpt).

> *Following your advice, the author put a tenth revision package together... this could be the basis of a first push towards a formalization by you. Take the time to review any changes, then if you think the basis is solid enough please go ahead and start developing the Lean code here.... The goal would be to isolate the desired end theorem in a file with as few imports, notation etc. as possible, with a clear certificate that the proof is* sorry- *and axiom-free.*

Reply (summary). Produced the bridge identifying C *with* $\exp(\sum c_r t^r)$*, the majorant inequality, the exact and dyadic-interval finite certificates, and the rational toolkit underlying the sign lock.*
Human action. Reviewed and integrated the layers, then handed the remaining analytic tail to a dedicated completion run.

Card 5: Long-horizon completion drive **OpenAI Codex CLI**

Task. Drive the formalization to completion with minimal supervision.
Prompt.

> */goal Please continue working on the Lean formalization of the argument, until it is either finished or until you hit a serious obstacle which requires either mathematical input or technical assistance beyond the standard difficulties and problem solving of the formalization process.*

Reply (summary). Over many cycles of inspection, coding, compilation, benchmarking, and repair, the worker proposed proof decompositions, searched Mathlib, refactored, generated finite-certificate checkers, and profiled slow computations.
Human action. Monitored progress, spot-checked the Lean state, and intervened only at the obstacle the goal itself anticipated (Card 6).
Technique. A persistent goal that doubles as a completion condition *attached to the thread [6]; before this, continuation had to be re-prompted every twenty to thirty minutes.*

Card 6: "Is the worker stuck?" — a review of the process **ChatGPT 5.5 Pro; Claude Opus 4.8**

Task. Decide whether the multi-day Codex run was healthy exploration or a dead end, and how to finish.
Prompt (excerpt).

> *Please review both the proof and the formalization files in detail. (i) Do you see any mathematical errors or obstructions in the strategy, and are they fixable? (ii) The formalization has run with a Codex worker for about two days; is this the expected process of investigation and optimization, or is the worker stuck in a local minimum or dead end? (iii) Any advice on how to finish, which could be forwarded to the worker.*

Reply (summary). Run in parallel in both systems; both concurred that the mathematics was sound but the worker was genuinely stuck — reshaping one product-tail reduction rather than proving it — and advised proving a single combined-product inequality directly instead.
Human action. Forwarded the advice to the worker and gathered the worker's own specific blocking questions for the deeper review of Card 7.
Technique. Reviewing the process, *not only the mathematics, and running the same review in two independent systems.*

Card 7: The correction that unstuck the proof **ChatGPT 5.5 Pro; Claude Opus 4.8**

Task. Resolve the worker's blocking questions about the large-tail product estimate.
Prompt (relayed, excerpt).

> *[Relaying the worker's own questions:] What is the most direct analytic proof of the sharp large-tail product target, preferably by modifying the existing split rather than introducing a large generated table?*

Reply (summary). The forwarded answer showed that the attempted route — bounding the two factors separately and multiplying — is not merely expensive but false *at the first large-tail cell* $(a, k) = (3000, 4)$*, off by about* $3.26 \cdot 10^{557}$ *because the split discards a factor* 2^{a-k}*; it prescribed the fix used in Section 6: keep the product together, bound it with finite exponential prefixes, prove the prefix-convolution inequality, and normalize only at the last step.*
Human action. Relayed the corrected route to the worker, which implemented it as the direct-saddle method; the proof closed shortly afterward.
Technique. Escalating the worker's own *questions to a fresh, stronger review, and letting a provably false target redirect the architecture.*

What the two interventions show. The episode in Cards 6 and 7 is worth keeping in the record. The autonomous worker had made genuine local progress around an interface that was globally false: separating the two factors and multiplying them loses the dyadic cancellation, and no amount of finite refinement could have repaired it. What exposed this was not a sharper estimate but the formal target itself, which refused to close. Formal verification thus did more than confirm a known proof: it prevented a plausible but invalid analytic architecture from being mistaken for a theorem, and it marked exactly the moment at which the autonomous worker needed an outside perspective. That perspective, too, came from an AI model; the author's part was the orchestration — reading a high-level signal that the run had stalled and routing the question to a fresh review — not any judgement about the mathematics. The result is autonomous in execution, and the decisions that mattered were about *when* the worker was stuck, not how to fix it.

Scope of this record. The seven cards are a curated, not exhaustive, account: they record the interactions that materially shaped the proof and omit many routine continuation prompts and unproductive sessions.

The write-up itself was likewise AI-produced, in three stages. The original technical research note — the tenth revision, focused purely on the mathematics — was authored by ChatGPT 5.5 Pro in the initial research strand of Card 1. After the formalization was complete, a fresh expository draft of the present paper was produced by ChatGPT 5.5 Pro in a new conversation, following the author's high-level instructions on organization and intended audience. The subsequent revisions — including those prompted by an external review of the release candidate — were carried out in Claude Code under the author's direction, together with further ChatGPT 5.5 Pro review. As with the mathematics, the author's contribution to the write-up was direction and editorial judgement rather than composition; and acceptance of each mathematical claim ultimately rested on its Lean formalization, not on any model's assurance.

Quantitative notes. The "roughly sixty-four hours" attributed to the completion phase is the elapsed wall-clock time recorded in the Codex goal log for that thread; it is not active compute time, is not summed over parallel workers, and includes waiting and compilation. Human active time was not separately measured, and token counts and monetary cost were not retained. The figure is given only to convey scale. The formal proof was checked under Lean 4.27.0 and a Mathlib revision pinned in the package manifest.

On the persistent goal. The `/goal` prompt of Card 5 is reproduced as it was actually used. By later best-practice standards [6] it is deliberately minimal: it states the desired outcome and the condition under which to stop and ask for help, but not the verification command, the permitted files, or an iteration policy. In practice the evolving Lean goal state, successful compilation, and the axiom reports served as the implicit evidence surface; a reproduction attempt should make those conditions explicit.

Limitations. Formal verification establishes the formal theorem relative to the trust boundary of Section 9. It does not by itself establish novelty or significance, the correctness of bibliographic claims, or the faithfulness of every sentence of the informal paper to the formal definitions; those remain matters of human and community judgement. The development also illustrates characteristic failure modes of AI-assisted research, which this record deliberately preserves: plausible but false intermediate estimates, a proliferation of proof interfaces, and convincing local progress on a globally false target. Curated, redacted transcripts of the command-line-agent sessions — every genuine human prompt, with the AI side summarized — together with the most relevant review exchanges are archived under `docs/ai-correspondence/`, and the card titles link to the primary public conversations, so that the provenance can be inspected directly.

11 Reproducibility and repository organization

The repository exposes a small public API:

`Statement.lean`, `Theorem.lean`, `Prop51.lean`.

The file `Theorem.lean` is intended as the first audit surface: it names the series `C`, states the quotient-series identity, and exposes the strict-negativity and non-vanishing theorems. The root file `Prop51.lean` imports this facade.

The proof backend remains more historical. In the current release, `Completion.lean` imports `OpenGoals.lean`, which contains both the live direct-saddle assembly and older certificate interfaces retained for audit and development history. This does not affect the public theorem or its axiom report, but a future presentation-only refactor can split the live direct-saddle construction into a small `Assembly.lean` and move the historical interfaces to a legacy namespace. Tagged releases in the repository preserve both the completed proof state and the public-facade state.

The release records the Lean toolchain, Mathlib revision, generated certificate provenance, and build commands. Continuous integration is set to run on both `main` and `master`; the Lake manifest agrees with the package name; the compact public axiom report is run in CI; and internal AI correspondence is kept in a development archive rather than at repository root.

A Public Lean statement

The following is a Lean-style rendering of the public facade, lightly edited for readability (e.g. `Q[[X]]` for the rational power-series type and ASCII `<=`, `!=` for the order and disequality); it is not literal source. The exact, machine-checked file is `Prop51/Theorem.lean` in the repository. It is reproduced to make the correspondence between the paper and the formal theorem immediate.

```
import Prop51.BCoeffSeries
import Prop51.Completion

namespace Prop51

open PowerSeries

noncomputable abbrev chenLarsonC : Q[[X]] := Cseries

theorem coeff_chenLarsonC (k : Nat) :
    coeff k chenLarsonC =
      (Nat.factorial (6*k) : Q) /
        ((Nat.factorial (3*k) : Q) *
          (Nat.factorial (2*k) : Q) * 72^k)

noncomputable abbrev chenLarsonSeries (mu : List Nat) : Q[[X]] :=
  bSeries mu

theorem chenLarsonSeries_spec (mu : List Nat) :
    chenLarsonC ^ ((mu.map (fun m => m + 1)).sum) *
      chenLarsonSeries mu
      = (mu.map (fun m =>
          rescale (((m + 1 : Nat) : Q)^(-1)) chenLarsonC)).prod
```

```
theorem chenLarsonCoefficient_neg
    {g : Nat} (hg : 2 <= g) (hmod : g % 3 != 1)
    {mu : List Nat} (hmu : IsPartitionOf mu (2 * g - 2)) :
    coeff (g / 3 + 1) (chenLarsonSeries mu) < 0

theorem chenLarsonCoefficient_ne_zero
    {g : Nat} (hg : 2 <= g) (hmod : g % 3 != 1)
    {mu : List Nat} (hmu : IsPartitionOf mu (2 * g - 2)) :
    coeff (g / 3 + 1) (chenLarsonSeries mu) != 0

end Prop51
```

B Paper-to-Lean correspondence

paper item	principal Lean declaration or file
hypergeometric series C	`Aseq`, `Cseries` in `ExpSeries.lean`
$\mathsf{C} = \exp(\sum c_r X^r)$	`Cseries_eq_expSeries_c` in `Bridge.lean`
quotient-series identity	`bSeries_official` in `BCoeffSeries.lean`
partition-free majorant	`bCoeff_le_U` in `Majorant.lean`
coefficient bounds	`d_lb`, `d_ub`, `d_increment`, `d_ratio_lb` in `DNorm.lean`
sign lock	`Xnorm_le_neg_final_margin` in `SignLock.lean`
finite saddle and prefix convolution	`expCoeff_saddle`, `expPrefix_mul_le` in `SaddleDirect.lean`
direct product estimates	`Bq_Qq_small_core`, `Bq_Qq_tempered_core` in `SaddleDirect.lean`
finite edge budget	`positiveSaddleFiniteScaledEdgeBudget` in `Generated/PositiveSaddleFiniteScaledEdge.lean`
closed proof	`completion_of_directSaddle_closed` and `coefficientNegativity`
public theorem	`chenLarsonCoefficient_neg`, `chenLarsonCoefficient_ne_zero` in `Theorem.lean`

C Computational certificate inventory

range	method	verification surface
$a \le 8$	exact partition enumeration	exact rational coefficient for every positive partition; selected partition-cardinality checks
$9 \le a \le 60$	exact rational majorant	all 10764 rectangle pairs
$61 \le a \le 400$	192-bit outward dyadic intervals	proved enclosure of recurrence tables; eight native chunks
$401 \le a \le 2000$	direct saddle plus fixed-point edge certificate	sixteen 100-row shards, scale 10^{20}; analytic solo theorem
$2001 \le a < 3000$	direct saddle plus generated prefix ratios	pointwise product and solo bounds; finite quotient/raw-budget chunks
$a \ge 3000$	symbolic rational tail	adjacent quotients, reverse upper band, unit reserves, and sharp solo envelope

References

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